

Differential Equations

Labix

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Abstract

These notes will act as an introductory text with a collection of theorems and definitions for differential equations.

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1.1

Definition 1.1.1 (Partial Differential Equations)

Let $U \subseteq \mathbb{R}^n$ be an open subset. A partial differential equation on U is an equation of the form

$$F(x, u(x), Du(x), \dots, D^k u(x)) = 0$$

where $u : U \rightarrow \mathbb{R}$ is a C^k -function and $F : U \times \mathbb{R} \times \mathbb{R}^n \times \dots \times \mathbb{R}^{n^k}$ is a function.

Definition 1.1.2 (Linear Types of Partial Differential Equations)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $u : U \rightarrow \mathbb{R}$ be a C^k -function. Let $F : U \times \mathbb{R} \times \mathbb{R}^n \times \dots \times \mathbb{R}^{n^k}$ be a partial differential equation on u .

- We say that F is linear if F is of the form

$$F = \sum_{i=0}^k p_i D^i u - f$$

where $p_i : U \rightarrow \mathbb{R}$ for $0 \leq i \leq k$ and f are functions.

Definition 1.1.3 (Homogeneous Partial Differential Equations)

Let $U \subseteq \mathbb{R}^n$ be an open subset. Let $u : U \rightarrow \mathbb{R}$ be a C^k -function. Let $F : U \times \mathbb{R} \times \mathbb{R}^n \times \dots \times \mathbb{R}^{n^k}$ be a partial differential equation on u . We say that F is homogeneous if

2 Four Important Linear Partial Differential Equations

2.1 Transport Equation

Definition 2.1.1 (Linear Transport Equation)

Define the transport equation to be the partial differential equation

$$\frac{\partial u}{\partial t} + b \cdot \left(\frac{\partial u}{\partial x_1}, \dots, \frac{\partial u}{\partial x_n} \right) = f$$

for some $b \in \mathbb{R}^n$ and some function $f : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$, and $u : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$ is the solution where we use the convention $u = u(x_1, \dots, x_n, t)$.

Definition 2.1.2 (The Initial Value Problem for Linear Transport Equations)

The initial value problem for the linear transport equation consists of finding a function $u : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$ such that the following are true.

- u is the solution to the transport equation:

$$\frac{\partial u(x, t)}{\partial t} + b \cdot \left(\frac{\partial u(x, t)}{\partial x_1}, \dots, \frac{\partial u(x, t)}{\partial x_n} \right) = f(x, t)$$

for some $b \in \mathbb{R}^n$ and some function $f : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$.

- $u(-, 0) = g$.

Proposition 2.1.3

Let $b \in \mathbb{R}^n$. Let $f : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$ be an integrable function. Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function. Then the function $u : \mathbb{R}^n \times (0, \infty) \rightarrow \mathbb{R}$ defined by

$$u(x, t) = g(x - tb) + \int_0^t f(x + (s - t)b, s) ds$$

is a solution to the initial value problem for linear transport equations.

2.2 Wave Equation

2.3 Heat Equation

2.4 Laplace's Equation

3 Introduction to Differential Equations

3.1 Classification

Definition 3.1.1 (Dependency) We say that a variable x is dependent on y if $y = f(x)$ for some function f , otherwise it is independent.

Definition 3.1.2 (Differential Equations) A differential equation is a function of variables and their derivatives.

Definition 3.1.3 (Ordinary and Partial) A differential equation is said to be ordinary if there is only one independent variable. It is partial otherwise. In particular,

- The equation

$$\left(y \frac{dy}{dx} + x \right) = \frac{d^2 y}{dx^2}$$

is ordinary

- The equation

$$\frac{du}{dx} \left(\frac{d^2u}{dx^2} + \frac{d^2u}{dt^2} \right) = u$$

is partial

Definition 3.1.4 (Autonomous and Non-autonomous) A differential equation is said to be autonomous if the independent variable does not appear explicitly. It is non-autonomous otherwise. In particular,

- The equation

$$y'' + 2y' = y'''y + 4$$

is autonomous

- The equation

$$\frac{dy}{dx} + \frac{d^2y}{dx^2} = x$$

is non-autonomous

Definition 3.1.5 (Linear and non-linear) An ordinary differential equation with y dependent on x is said to be linear if the coefficient of every derivative is a function of x or a constant. It is non-linear otherwise. In particular,

- The equation

$$x^2 \frac{dy}{dx} = 6 \frac{d^2y}{dx^2}$$

is linear

- The equation

$$y \frac{dy}{dx} = \left(\frac{d^2y}{dx^2} \right)^2$$

is non-linear

Definition 3.1.6 (Homogenous and non-homogenous) An ordinary differential equation is said to be homogenous if the term without a derivative is 0. In particular,

- The equation

$$y'' + 2y' + y = 0$$

is homogenous

- The equation

$$y''' + yy' = xy + \sin(x)$$

is not homogenous

4 Ordinary Differential Equations: First Order

4.1 Existence and Uniqueness of Solutions of First Order Differential Equations

Theorem 4.1.1 [Globally Lipschitz Picard's Theorem] Let $f : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ is continuous and that there exists $L > 0$ such that

$$|f(x, t) - f(y, t)| \leq L|x - y|$$

for all $x, y \in \mathbb{R}^n, t \in \mathbb{R}$. If $\tau L < 1$, then for every $x_0 \in \mathbb{R}^n$, there exists a unique differentiable function $x : [t_0, t_0 + \tau] \rightarrow \mathbb{R}^n$ that satisfies

$$\frac{dx}{dt} = f(x, t)$$

with $x(t_0) = x_0$ for all $t \in [t_0, t_0 + \tau]$.

Proposition 4.1.2 Suppose that a solution is guaranteed by the above theorem on $[t_0, t_1]$ given by $x_1(t)$ and another solution is guaranteed on $[t_1, t_2]$ given by $x_2(t)$. Then

$$x(t) = \begin{cases} x_1(t) & t_0 \leq t \leq t_1 \\ x_2(t) & t_1 \leq t \leq t_2 \end{cases}$$

is a solution for the differential equation $x'(t) = f(x, t)$ on $[t_0, t_2]$

Theorem 4.1.3 Suppose we have a system of linear differential equations $\mathbf{x}'(t) = A\mathbf{x}(t)$ with initial condition $x(0) = x_0$ where A is an $n \times n$ matrix. Then the solution of this is

$$x(t) = e^{tA}x_0 = \sum_{k=0}^{\infty} \frac{t^k A^k}{k!} x_0$$

for $|t| \leq \tau$ where $\tau < \|A\|^{-1}$

Theorem 4.1.4 [Locally Lipschitz Picard's Theorem] Let $f : \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ is continuous and that there exists $L > 0$ such that

$$|f(x, t) - f(y, t)| \leq L|x - y|$$

for all $x, y \in \mathbb{R}^n, t \in \mathbb{R}$. If $\tau L < 1$, then for every $x_0 \in \mathbb{R}^n$, there exists a unique differentiable function $x : [t_0, t_0 + \tau] \rightarrow \mathbb{R}^n$ that satisfies

$$\frac{dx}{dt} = f(x, t)$$

with $x(t_0) = x_0$ for all $t \in [t_0, t_0 + \tau]$.

Theorem 4.1.5 Suppose that $U \subset \mathbb{R}^n$ is open and that $f : U \rightarrow \mathbb{R}^n$ is continuous and locally Lipschitz, meaning for every compact subset K of U there exists L_K such that

$$|f(u) - f(v)| \leq L_K|u - v|$$

for $u, v \in K$, then for some $\tau = \tau(x_0) > 0$ there exists a unique $x : [t_0 - \tau, t_0 + \tau] \rightarrow U$ that solves

$$\frac{dx}{dt} = f(x)$$

with initial condition $x(t_0) = x_0$ for all $t \in [t_0 - \tau, t_0 + \tau]$.

4.2 Separable Equations

Definition 4.2.1 (Separable Equations) If $\frac{dx}{dt} = f(x)g(t)$. Then it is called separable.

Lemma 4.2.2 A separable differential equation $\frac{dx}{dt} = f(x)g(t)$ can be solved by solving

$$\int \frac{1}{f(x)} dx = \int g(t) dt$$

4.3 The Linear Case

Theorem 4.3.1 [First Order Linear Equation] Let $R(t) = \int r(t) dt$. The differential equation

$$\frac{dx}{dt} + r(t)x = g(t)$$

has the solution

$$x(t) = Ae^{-R(t)} + e^{-R(t)} \int e^{R(t)} g(t) dt$$

Proof We first multiply both sides of the equation by $e^{R(t)}$. We have

$$\begin{aligned} \frac{dx}{dt} + r(t)x &= g(t) \\ e^{R(t)} \frac{dx}{dt} + r(t)e^{R(t)}x &= g(t)e^{R(t)} && \text{(Multiply by } e^{R(t)}) \\ \frac{d}{dt} (e^{R(t)}x) &= g(t)e^{R(t)} \\ e^{R(t)}x + C &= \int g(t)e^{R(t)} dt && \text{(By the FTC)} \\ x &= e^{-R(t)} \int g(t)e^{R(t)} dt + Ae^{-R(t)} && \text{(Relabel } A = -C) \end{aligned}$$

Corollary 4.3.2 If $r(t) = p$ is a constant function and $g(t) = 0$, then the differential equation

$$\frac{dx}{dt} + r(t)x = g(t)$$

has the solution

$$x(t) = Ae^{-\int r(t) dt}$$

Corollary 4.3.3 If $r(t) = p$ is a constant function, then the differential equation

$$\frac{dx}{dt} + r(t)x = g(t)$$

has the solution

$$x(t) = Ae^{-pt}$$

4.4 Substitution Method

Theorem 4.4.1 (Type 1 Substitution) Consider differential equations of the form

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Then let $u = \frac{y}{x}$ to obtain

$$x \frac{du}{dx} = F(u) - u$$

Theorem 4.4.2 (Type 2 Substitution) Consider differential equations of the form

$$\frac{dy}{dx} + p(x)y = q(x)y^n$$

Then let $u = y^{1-n}$ to obtain

$$\frac{du}{dx} + (1-n)p(x)u = (1-n)q(x)$$

Proposition 4.4.3 For a differential equation in the form $f(x, y', y'') = 0$, the equation can be reduced to a first order differential equation by substitution.

Proposition 4.4.4 For a differential equation in the form $f(y, y', y'') = 0$, the equation can be reduced to a first order differential equation by substitution

4.5 Systems of First Order Equations

Definition 4.5.1 [System of First Order Equations] A system of first order equations is given by

$$\mathbf{x}'(t) = A\mathbf{x}$$

where $\mathbf{x}'(t) = \begin{pmatrix} \frac{dx_1}{dt} \\ \vdots \\ \frac{dx_n}{dt} \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$

Theorem 4.5.2 Consider a system of first order equation with constant coefficients

$$\begin{cases} \frac{dx}{dt} = px + qy \\ \frac{dy}{dt} = rx + sy \end{cases}$$

with matrix equation

$$\mathbf{x}'(t) = A\mathbf{x}$$

Suppose A has eigenvalues $\lambda_{1,2}$ and eigenvectors $\mathbf{v}_{1,2}$.

- If $\lambda_{1,2}$ are distinct, then

$$\mathbf{x}(t) = e^{\lambda_1 t} \mathbf{v}_1 + e^{\lambda_2 t} \mathbf{v}_2$$

- If $\lambda_{1,2} \in \mathbb{C}$, let $\mathbf{v} = \mathbf{v}_1 + i\mathbf{v}_2$. Then

$$\mathbf{x}(t) = \mathbf{c}\mathbf{v}e^{\lambda t} + \overline{\mathbf{c}\mathbf{v}}e^{\bar{\lambda}t} = 2\operatorname{Re}(\mathbf{c}\mathbf{v}e^{\lambda t})$$

where c is such that $\mathbf{x}(t)$ is real. Or

$$\mathbf{x}(t) = e^{pt}[a \cos(qt) + b \sin(qt)\mathbf{v}_1 + (b \cos(qt) - a \sin(qt))\mathbf{v}_2]$$

- If the eigenvalues are repeated, then $\mathbf{x}(t) = \mathbf{a}e^{\lambda t} + \mathbf{b}te^{\lambda t}$

Theorem 4.5.3 [Uncoupling Solutions] Let $\dot{\mathbf{x}} = A\mathbf{x}$ be a system of differential equations. Let it has distinct eigenvalues. Then there exists $P = (\mathbf{v}_1 | \mathbf{v}_2)$ such that $P^{-1}AP = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$ Then let

$$\dot{\mathbf{y}} = P^{-1}\dot{\mathbf{x}}. \text{ Then } \dot{\mathbf{y}} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \mathbf{y}$$

4.6 Direction Fields

Definition 4.6.1 [Vector Fields] A planar vector field is a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Definition 4.6.2 [Direction Fields] Let $\frac{dx}{dt} = f(x, t)$ be a differential equation. A direction field is vector field $v : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that each vector $(x, y) \in \mathbb{R}^2$ is assigned the vector with slope $f(x, t)$.

Definition 4.6.3 [Fixed Points] We say that $(x_0, f(x_0))$ is a fixed point if $\frac{dx}{dt} = f(x) = 0$. A fixed point is stable if $f'(x_0) < 0$. It is unstable if $f'(x_0) > 0$.

5 Ordinary Differential Equations: Second Order

5.1 Second Order Linear Ordinal Differential Equations

Definition 5.1.1 [Linearly Independent Solutions] Two functions $x_1(t), x_2(t)$ defined on an interval I are linearly independent if $\alpha_1 x_1(t) + \alpha_2 x_2(t) = 0 \implies \alpha_1 = \alpha_2 = 0$.

Definition 5.1.2 The complete solution to a inhomogeneous solution is given by $x(t) = x_h(t) + x_p(t)$ where $x_h(t)$ is the solution to the homogeneous solution and $x_p(t)$ is the particular solution to the equation.

Definition 5.1.3 [Auxiliary Equations] Consider

$$a \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + cx = 0$$

The auxiliary equation of the differential is $ak^2 + bk + c = 0$.

Theorem 5.1.4 Consider

$$a \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + cx = 0$$

and $k_{1,2}$ roots of $ak^2 + bk + c = 0$

- If $b^2 - 4ac > 0$, then $x(t) = Ae^{k_1 t} + Be^{k_2 t}$
- If $b^2 - 4ac = 0$, then $x(t) = (A + Bt)e^{kt}$
- If $b^2 - 4ac < 0$, then $x(t) = e^{pt}(C \cos(qt) + D \sin(qt))$, where $k_{1,2} = p \pm iq$ or $x(t) = Ee^{pt} \cos(qt - \phi)$ where $E^2 = C^2 + D^2$ and $\tan(\phi) = \frac{D}{C}$

Theorem 5.1.5 Let

$$a \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + cx = f(t)$$

To find the particular solution, plug the possibility into the differential to find identities for coefficients.

- If $f(t)$ is a polynomial, then $x_p(t)$ is a general polynomial of degree n .
- If $f(t) = ae^{kt}$ where k is not a root of the auxiliary equation, then $x_p(t) = Ae^{kt}$
- If $f(t) = ae^{kt}$ where k is a root of the auxiliary equation, then $x_p(t) = Ate^{kt}$ or At^2e^{kt}
- If $f(t) = a \sin(\omega t)$ or $f(t) = a \cos(\omega t)$ then $x_p(t) = A \sin(\omega t) + B \cos(\omega t)$
- If $f(t) = at^n e^{kt}$ then $x_p(t) = P(t)e^{kt}$, where $P(t)$ is a polynomial.
- If $f(t) = t^n(a \sin(\omega t) + b \cos(\omega t))$ then $x_p(t) = P_1(t)a \sin(\omega t) + P_2(t)b \cos(\omega t)$
- If $f(t) = e^{kt}(a \sin(\omega t) + b \cos(\omega t))$ then $x_p(t) = e^{kt}(A \sin(\omega t) + B \cos(\omega t))$

6 Difference Equations

6.1 First Order Homogeneous Linear Difference Equations

Definition 6.1.1 (Order of a Differential Equation) The order of a difference equation is the difference between the highest index of x and lowest.

Definition 6.1.2 (First Order Difference Equations) First order difference equations are of the form

$$x_{n+1} = f(x, n)$$

Theorem 6.1.3 The solution to a first order homogeneous linear difference equation

$$x_{n+1} = ax_n$$

is given by $x_n = a^n x_0$.

Definition 6.1.4 (Fixed Point) A fixed point of the difference equation $x_{n+1} = f(x_n)$ is a point x_0 such that $f(x_0) = x_0$

Proposition 6.1.5 (Stability of Fixed Points) If $|a| < 1$, then for any x_0 , x_0 is stable. If $|a| > 1$, then x_0 is unstable.

6.2 Second Order Linear Difference Equations

Theorem 6.2.1 Consider an equation of the form

$$x_{n+2} + ax_{n+1} + bx_n = 0$$

Let $k_{1,2}$ be roots of the equation $k^2 + ak + b = 0$

- If $k_{1,2}$ are distinct, then the general solution is $x_n = Ak_1^n + Bk_2^n$
- If $k_1 = k_2$, then the general solution is $Ak_1^n + Bnk_1^n$
- If $k_1, k_2 \in \mathbb{C}$, then let $k_{1,2} = k_{\pm} = re^{\pm i\theta}$. Then the general solution is $x_n = r^n (A \cos(n\theta) + B \sin(n\theta))$

Theorem 6.2.2 Consider an equation of the form

$$x_{n+2} + ax_{n+1} + bx_n = f(n)$$

Let $k_{1,2}$ be roots of the equation $k^2 + ak + b = 0$.

- If $f(n)$ is a polynomial, try a polynomial.
- If $f(n) = a^n$. Try $x_n = Ca^n$ or Cna^n or Cn^2a^n

7 Partial Differential Equations: Notations

7.1 Terminology

Definition 7.1.1 (Partial Differential Equations) A partial differential equations is an equation of the form

$$F(x_1, \dots, x_n, u, \partial_{x_1} u, \dots, \partial_{x_n} u, \partial_{x_1 x_1} u, \dots, \partial_{x_n x_n} u, \dots) = 0$$

We some times use the differential operator $\mathcal{L}$ to replace the differentials of the equation. The equation thus becomes

$$\mathcal{L}(u) = r(x_1, \dots, x_n)$$

where $r(x_1, \dots, x_n)$ is a function only of the independent variables.

Definition 7.1.2 (Classification of Partial Differential Equations) We say that a partial differential equation $\mathcal{L}(u) = r(x)$ is

- linear if the differential operator is linear, meaning $\mathcal{L}(au + bv) = a\mathcal{L}(u) + b\mathcal{L}(v)$ for $a, b \in \mathbb{R}$ and u, v functions
- homogenous if $r(x) = 0$, meaning $\mathcal{L}(u) = 0$
- of order n if the highest order of possibly mixed partial derivative of u is n

7.2 Well-Posedness

Definition 7.2.1 (Hadmark's Well-posedness) A partial differential equation problem is well-posed if it has a unique solution that continuously depends on the data. In other words, we want existence, uniqueness and stability in the solution.

7.3 Classification of Linear Second Order PDEs

Definition 7.3.1 (Classification of Linear Second Order PDEs) Let $\sum_{i=1}^d \sum_{j=1}^d a_{ij}(x) \partial_{x_i x_j} u(x) + b(x) = 0$ where $b(x)$ consists of lower order terms involving derivatives up to order 1. Define the coefficient matrix A by

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$$

for $x \in U \subseteq \mathbb{R}^d$. We say that

- A is elliptic for $x \in U$ if A has d eigenvalues of the same sign
- A is hyperbolic for $x \in U$ if A has $d - 1$ eigenvalues of the same sign and one eigenvalue of the opposite sign
- A is parabolic for $x \in U$ if one eigenvalue is 0 and the other $d - 1$ have the same sign

8 Partial Differential Equations: Transport Equation

8.1 Homogenous Transport Equation

Definition 8.1.1 (Transport Equation) The transport equation is the equation

$$\partial_t u(x, t) + v(x, t) \partial_x u(x, t) = 0$$

or

$$\frac{\partial u}{\partial t}(x, t) + v(x, t) \frac{\partial u}{\partial x}(x, t) = 0$$

Definition 8.1.2 (Characteristics) The characteristic is the function $\xi(t)$ satisfying the ODE

$$\xi'(t) = v(\xi(t), t)$$

where $\xi(0) = x_0 \in \mathbb{R}$.

Proposition 8.1.3 The solution to the transport equation is constant along the characteristics curve.

Proof We have that letting the position x depends on t , meaning $x = \xi(t)$,

$$\begin{aligned} \frac{d}{dt} u(\xi(t), t) &= \partial_x u(\xi(t), t) \xi'(t) + \partial_t u(x, t) \\ &= \partial_x u(\xi(t), t) v(x, t) + \partial_t u(x, t) \\ &= 0 \end{aligned}$$

Thus u is constant along the characteristics. ■

8.2 Inhomogenous Transport Equation

Definition 8.2.1 (Transport Equation with Source) The transport equation with source term is the equation

$$\partial_t u(x, t) + v(x, t) \partial_x u(x, t) = s(u)$$

or

$$\frac{\partial u}{\partial t}(x, t) + v(x, t) \frac{\partial u}{\partial x}(x, t) = s(u)$$

Proposition 8.2.2 Given the characteristic function $\xi(t)$, the transport equation with source amounts to solving the differential equation

$$\frac{d}{dt} u(\xi(t), t) = s(\xi(t), t, u(\xi(t), t))$$

with given initial conditions.

Proof If $u(x, t) = u(\xi(t), t)$, then

$$\begin{aligned} \frac{d}{dt} u(\xi(t), t) &= \xi'(t) \partial_x u(\xi(t), t) + \partial_t u(\xi(t), t) \\ &= v(\xi(t), t) \partial_x u(\xi(t), t) + \partial_t u(\xi(t), t) \\ &= s(\xi(t), t, u(\xi(t), t)) \end{aligned}$$

thus we are done. ■

8.3 Non-Linear Transport Equation

Definition 8.3.1 (Non-Linear Transport Equation) The non-linear transport equation is the equation

$$\partial_t u(x, t) + v(u) \partial_x u(x, t) = 0$$

or

$$\frac{\partial u}{\partial t}(x, t) + v(u) \frac{\partial u}{\partial x}(x, t) = 0$$

The difference being that v is now a function of u .

9 Partial Differential Equations: Wave Equation

9.1 Wave Equation

Definition 9.1.1 (Wave Equation in One Spatial Dimension) The wave equation in one dimension is defined to be

$$\partial_{tt}u = c^2 \partial_{xx}u$$

where c is the speed of the sound waves.

Proposition 9.1.2 The wave operator corresponds to twice applying the transport operator with opposite velocity signs.

Proof Note that

$$\begin{aligned} (\partial_t - c\partial_x)(\partial_t + c\partial_x)u &= (\partial_t - c\partial_x)(\partial_t u + c\partial_x u) \\ &= \partial_{tt}u - c\partial_{xt}u + c\partial_{tx}u - c^2\partial_{xx}u \\ &= \partial_{tt}u - c^2\partial_{xx}u \end{aligned}$$

By expanding, we also have that

$$(\partial_t + c\partial_x)(\partial_t - c\partial_x)u = \partial_{tt}u - c^2\partial_{xx}u$$

mainly since the partial derivatives are well defined and thus commute. ■

Theorem 9.1.3 The general solution of the wave equation is given by

$$u(x, t) = f(x + ct) + g(x - ct)$$

where $f, g : \mathbb{R} \rightarrow \mathbb{R}$, using the coordinate method.

Proof By finding suitable coordinates $\xi = x + ct$ and $\nu = x - ct$, we can linearly transform the equation in to one the only involves $\partial_{\xi\nu}$. In particular, letting $v(\xi, \nu) = u(x, t)$, we have

$$\partial_x u = \partial_\xi v + \partial_\nu v$$

and

$$\partial_t u = c\partial_\xi v - \partial_\nu v$$

so that

$$\partial_{xx}u = \partial_{\xi\xi}v + \partial_{\nu\xi}v + \partial_{\nu\xi}v + \partial_{\nu\nu}v$$

and

$$\partial_{tt}u = c^2\partial_{\xi\xi}v - c^2\partial_{\xi\nu}v - c^2\partial_{\nu\xi}v + c^2\partial_{\nu\nu}v$$

If u, v are smooth enough, we have that

$$\partial_{tt}u - c^2\partial_{xx}u = c^2(-4\partial_{\xi\nu}v)$$

and our question becomes

$$\partial_{\xi\nu}v = 0$$

Integrating the mixed partial derivative by each of its variables once results in the form

$$u(x, t) = f(x + ct) + g(x - ct)$$

9.2 Initial Value Problem on Wave Equations

Definition 9.2.1 (Cauchy's Initial Value Problem) Let $\Psi : \mathbb{R} \rightarrow \mathbb{R}$ and $V : \mathbb{R} \rightarrow \mathbb{R}$ be two arbitrary functions. The initial value problem for the wave equation in one dimension consists of finding a function $u : \mathbb{R} \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_{tt}u(x, t) = c^2 \partial_{xx}u(x, t) & (x, t) \in \mathbb{R} \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in \mathbb{R} \\ \partial_t u(x, 0) = V(x) & x \in \mathbb{R} \end{cases}$$

Proposition 9.2.2 (d'Alembert's Formula) The solution to Cauchy's Initial Value Problem is given by

$$u(x, t) = \frac{1}{2} (\Psi(x + ct) + \Psi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} V(r) dr$$

provided that $\Psi \in C^2(\mathbb{R})$ and $V \in C^1(\mathbb{R})$.

Proof Using the change of coordinates method, we know that $u(x, t) = f(x + ct) + g(x - ct)$ thus

$$\Psi(x) = u(x, 0) = f(x) + g(x)$$

and

$$\Psi'(x) = f'(x) + g'(x)$$

and

$$V(x) = \partial_t u(x, 0) = cf'(x) - cg'(x)$$

and

$$\frac{1}{c}V(x) = f'(x) - g'(x)$$

Adding and subtracting the identities give

$$f'(x) = \frac{1}{2}(\Psi'(x) + \frac{1}{c}V(x))$$

and

$$g'(x) = \frac{1}{2}(\Psi'(x) - \frac{1}{c}V(x))$$

Integrating the two expressions with respect to x gives

$$f(x) = \frac{1}{2}\Psi(x) + \frac{1}{2c} \int_0^x V(r) dr + C_f$$

and

$$g(x) = \frac{1}{2}\Psi(x) - \frac{1}{2c} \int_0^x V(r) dr + C_g$$

Since $f(x) + g(x) = \Psi(x)$, we see that $C_f + C_g = 0$. Thus

$$u(x, t) = \frac{1}{2} (\Psi(x + ct) + \Psi(x - ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} V(r) dr$$

■

Proposition 9.2.3 (Preservation in Energy) Let $u \in C^2(\mathbb{R} \times (0, \infty))$ be a solution to the initial value problem for the wave equation. Let

$$E_k(t) = \int_{\mathbb{R}} \frac{1}{2} (\partial_t u)^2(x, t) dt$$

denote kinetic energy and

$$E_p(t) = \int_{\mathbb{R}} \frac{1}{2} c^2 (\partial_x u)^2(x, t) dt$$

denote potential energy. If

- $E_{WE}(t) = E_k(t) + E_p(t)$ is continuously differentiable with respect to t
- $u(x, t)$ and its partial derivatives satisfy $u(x, t) \rightarrow 0$ as $x \rightarrow \pm\infty$ for all t
- All mixed partial derivatives of u of up to order 2 are integrable with respect to x over $\mathbb{R}$ for all t

Then the total energy $E_{WE}(t)$ is constant in time.

Proof We want to show that $E'_{WE}(t) = 0$. Thanks to the assumptions, the following integrals and its calculations are allowed. We have

$$\begin{aligned} E'_{WE}(t) &= \frac{d}{dt} \left(\int_{\mathbb{R}} \frac{1}{2} ((\partial_t u)^2 + c^2 (\partial_x u)^2) dx \right) \\ &= \int_{\mathbb{R}} (\partial_t u \partial_{tt} u + c^2 \partial_x u \partial_{xt} u) dx \\ &= \int_{\mathbb{R}} \partial_t u \partial_{tt} u dx + \int_{\mathbb{R}} c^2 \partial_x u \partial_x (\partial_t u) dx \\ &= \int_{\mathbb{R}} \partial_t u (\partial_{tt} u - c^2 \partial_{xx} u) dx + c^2 \left(\lim_{x \rightarrow \infty} (\partial_x u(x, t) \partial_t u(x, t)) - \lim_{x \rightarrow -\infty} (\partial_x u(x, t) \partial_t u(x, t)) \right) \\ &\hspace{15em} \text{(By parts)} \\ &= \int_{\mathbb{R}} (\partial_t u \cdot 0) dx \\ &= 0 \end{aligned}$$

thus we are done. ■

Theorem 9.2.4 (Uniqueness and Existence) Let $\Psi \in C^2(\mathbb{R})$ and $V \in C^1(\mathbb{R})$. Then there is a unique solution $u \in C^2(\mathbb{R} \times (0, \infty))$ to the initial value problem for the one dimensional wave equation, given by d'Alembert's formula.

Proof Let u_1, u_2 be two solutions and $w = u_1 - u_2$. Then w is a solution of the initial value problem where

$$\partial_{tt} w(x, t) = c^2 \partial_{xx} w(x, t)$$

where $(x, t) \in (0, \infty) \times \mathbb{R}$, $w(x, 0) = 0$ for $x \in \mathbb{R}$ and $\partial_t w(x, 0) = 0$ for $x \in \mathbb{R}$. Thus the corresponding energy

$$E_{WE}(t) = \frac{1}{2} \int_{\mathbb{R}} ((\partial_t w)^2(x, t) + c^2 (\partial_x w)^2(x, t)) dx$$

is constant. Note that

$$E_{WE}(0) = \frac{1}{2} \int_{\mathbb{R}} ((\partial_t w)^2(x, 0) + c^2 (\partial_x w)^2(x, 0)) dx = 0$$

by the initial values. Thus we have that $E_{WE}(t) = E_{WE}(0) = 0$. Thus we have that

$$\partial_t w(x, t) = 0$$

and

$$\partial_x w(x, t) = 0$$

for all (x, t) . This means that $w(x, t)$ is constant. But $w(x, 0)$ thus $0 = u_1 - u_2$ and $u_1 = u_2$ and we are done. ■

Definition 9.2.5 (Cauchy's Initial Value Problem on Finite Spatial Domain) Let $\Psi : \mathbb{R} \rightarrow \mathbb{R}$ and $V : \mathbb{R} \rightarrow \mathbb{R}$ be two arbitrary functions. The initial value problem for the wave equation in one dimension consists of finding a function $u : (0, L) \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_{tt}u(x, t) = c^2 \partial_{xx}u(x, t) & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in (0, L) \\ \partial_t u(x, 0) = V(x) & x \in (0, L) \end{cases}$$

Proposition 9.2.6 (Change of Variables) Let $\Psi : \mathbb{R} \rightarrow \mathbb{R}$ and $V : \mathbb{R} \rightarrow \mathbb{R}$ be two arbitrary functions. The initial value problem for the wave equation in one dimension on finite spatial domain is equivalent to solving

$$\begin{cases} \partial_{tt}u(x, t) = \partial_{xx}u(x, t) & (x, t) \in (0, \pi) \times (0, \infty) \\ u(x, 0) = \hat{\Psi}(x) = \Psi\left(\frac{L}{\pi}x\right) & x \in (0, \pi) \\ \partial_t u(x, 0) = \hat{V}(x) = \frac{L}{\pi c}V\left(\frac{L}{\pi}x\right) & x \in (0, \pi) \end{cases}$$

9.3 Boundary Conditions on Wave Equations

Typical examples: motion of a string with fixed ends.

Definition 9.3.1 (Homogenous Dirichlet Boundary Condition) The initial boundary value problem for the wave equation in 1D with homogenous Dirichlet boundary condition consists of finding a function $u : (0, \pi) \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_{tt}u(x, t) = \partial_{xx}u(x, t) & (x, t) \in (0, \pi) \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in (0, \pi) \\ \partial_t u(x, 0) = V(x) & x \in (0, \pi) \\ u(0, t) = 0 & t \in (0, \infty) \\ u(\pi, t) = 0 & t \in (0, \infty) \end{cases}$$

Definition 9.3.2 (Homogenous Neumann Boundary Condition) The initial boundary value problem for the wave equation in 1D with homogenous Neumann boundary condition consists of finding a function $u : (0, \pi) \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_{tt}u(x, t) = \partial_{xx}u(x, t) & (x, t) \in (0, \pi) \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in (0, \pi) \\ \partial_t u(x, 0) = V(x) & x \in (0, \pi) \\ \partial_x u(0, t) = 0 & t \in [0, \infty) \\ \partial_x u(\pi, t) = 0 & t \in [0, \infty) \end{cases}$$

Proposition 9.3.3 Suppose that u_1, u_2 both solve either Neumann or Dirichlet's boundary conditions but with different initial data. Then $w = u_1 - u_2$ is bounded.

Lemma 9.3.4 Suppose that u_1, u_2 both solve either Neumann or Dirichlet's boundary conditions with the same initial data. Then $u_1 = u_2$.

Proposition 9.3.5 If $\Psi \in C^4(\mathbb{R})$ and $V \in C^3(\mathbb{R})$ and $\Psi(x) = \sum_{k=1}^{\infty} A_k \sin(kx)$ and $V(x) = \sum_{k=1}^{\infty} B_k k \sin(kx)$, then the solution to the Dirichlet Boundary Condition is given by

$$u(x, t) = \sum_{k=1}^{\infty} (A_k \cos(kt) + B_k \sin(kt)) \sin(kx)$$

Proof Assume that $u(x, t) = X(x)T(t)$ is separable where $X : (0, \pi) \rightarrow \mathbb{R}$ and $T : (0, \infty) \rightarrow \mathbb{R}$. Then the wave equation changes to

$$\frac{T''(t)}{T(t)} = \frac{X''(x)}{X(x)}$$

This ratio can only be constant since it cannot depend on t nor x . Let the ratio be λ . Then we now have two ODEs

$$X''(x) - \lambda X(x) = 0$$

and

$$T''(t) - \lambda T(t) = 0$$

I claim that $\lambda \leq 0$. Indeed since integrating any of the above two expression with respect to x from 0 to π gives

$$\begin{aligned} \int_0^\pi (-X''(x)X(x) + \lambda|X(x)|^2) dx &= 0 \\ \int_0^\pi \lambda|X(x)|^2 dx - [X'(x)X(x)]_0^\pi + \int_0^\pi |X'(x)|^2 dx & \\ \int_0^\pi |X'(x)|^2 dx + \lambda \int_0^\pi |X(x)|^2 dx & \quad (X(\pi) = X(0) = 0) \end{aligned}$$

If $\lambda > 0$ then $X(x) = 0$ for all $x \in (0, \pi)$ which is trivial. If $\lambda = 0$ then $X'(x) = 0$ and $X(x)$ is a constant function. But $X(0) = 0$ thus $X(x)$ is also identically zero.

Thus we assume that $\lambda = -\beta^2$ for some $\beta \in \mathbb{R}^+$. The solution to this ODE is precisely

$$X(x) = C \cos(\beta x) + D \sin(\beta x)$$

and

$$T(t) = A \cos(\beta t) + B \sin(\beta t)$$

From $X(0) = 0$ we have that $C = 0$ and $D \neq 0$. $X(\pi) = 0$ implies that $\beta\pi = k\pi$ for $k \in \mathbb{N} \setminus \{0\}$. Taking into account $T(t)$, we now have that

$$u_j(x, t) = (A_j \cos(kt) + B_j \sin(kt)) \sin(kx)$$

for $A_k, B_k \in \mathbb{R}$. Since any $k \in \mathbb{N}$ could solve the wave equation, we sum all the frequencies to get

$$u(x, t) = \sum_{k=1}^{\infty} (A_k \cos(kt) + B_k \sin(kt)) \sin(kx)$$

■

Proposition 9.3.6 If $\Psi \in C^4(\mathbb{R})$ and $V \in C^3(\mathbb{R})$ and $\Psi(x) = \sum_{k=1}^{\infty} A_k \cos(kx)$ and $V(x) = \sum_{k=1}^{\infty} B_k k \cos(kx)$, then the solution to the Dirichlet Boundary Condition is given by

$$u(x, t) = \sum_{k=1}^{\infty} (A_k \cos(kt) + B_k \sin(kt)) \sin(kx)$$

Proof We argue similarly by assuming separation of variables and take the ratio constant to be $\lambda = -\beta^2$ for some $\beta \in \mathbb{R}^+$. The solution to this ODE is precisely

$$X(x) = C \cos(\beta x) + D \sin(\beta x)$$

and

$$T(t) = A \cos(\beta t) + B \sin(\beta t)$$

From $X'(0) = 0$ we have that $D = 0$ and $C \neq 0$. $X'(\pi) = 0$ implies that $\beta\pi = k\pi$ for $k \in \mathbb{N}$. Taking into account $T(t)$, we now have that

$$u_j(x, t) = (A_j \cos(kt) + B_j \sin(kt)) \cos(kx)$$

for $A_k, B_k \in \mathbb{R}$. Since any $k \in \mathbb{N}$ could solve the wave equation, we sum all the frequencies to get

$$u(x, t) = \sum_{k=0}^{\infty} (A_k \cos(kt) + B_k \sin(kt)) \cos(kx)$$



10 Partial Differential Equations: Heat Equation

10.1 Heat Equation

Definition 10.1.1 (One Dimensional Heat Equation) The one dimensional heat equation is the equation

$$\partial_t u = k \partial_{xx} u$$

for a function $u(x, t)$ where $k > 0$.

10.2 Heat Equation: Dirichlet Boundary Conditions

Definition 10.2.1 (Dirichlet Boundary Conditions) The initial boundary value problem for the heat equation in one dimension with inhomogenous Dirichlet boundary condition consists of finding a function $u : (0, L) \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_t u(x, t) = k \partial_{xx} u(x, t) & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in [0, L] \\ u(0, t) = g_0(t) & t \in [0, \infty) \\ u(L, t) = g_L(t) & t \in [0, \infty) \end{cases}$$

Definition 10.2.2 (Space-Time Boundaries) We denote the space time rectangle by

$$V_{L,T} = (0, L) \times (0, T]$$

and the parabolic boundary by

$$\Gamma_{L,T} = \{(x, t) \in \overline{V_{L,T}} | x \in \{0, L\} \text{ or } t = 0\}$$

Theorem 10.2.3 (Maximum Principle) Assume that $u(x, t) \in C^2(\overline{V_{L,T}})$ solves the heat equation. Then the maximum and minimum of u are attained on the parabolic boundary $\Gamma_{L,T}$

Theorem 10.2.4 (Uniqueness and Stability) Let $u_1, u_2 \in C^2([0, L] \times [0, \infty))$ denote two solutions to the initial boundary value problem with respective initial data Ψ_1, Ψ_2 and boundary data $g_{1,0}, g_{1,L}, g_{2,0}, g_{2,L}$. Then

$$\max_{(x,t) \in V_{L,T}} |u_1(x, t) - u_2(x, t)| \leq \max\{A, B, C\}$$

where

- $A = \max_{x \in [0, L]} |\Psi_1(x) - \Psi_2(x)|$
- $B = \max_{t \in [0, T]} |g_{1,0} - g_{2,0}|$
- $C = \max_{t \in [0, T]} |g_{1,L} - g_{2,L}|$

In particular, if $\Psi_1 = \Psi_2, g_{1,0} = g_{2,0}$ and $g_{1,L} = g_{2,L}$ then $u_1 = u_2$.

10.3 Heat Equation: Neumann Boundary Condition

Definition 10.3.1 (Inhomogenous Neumann Boundary Condition) The initial boundary value problem for the heat equation in one dimension with inhomogenous Neumann boundary condition consists of finding a function $u : (0, L) \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_t u(x, t) = k \partial_{xx} u(x, t) & (x, t) \in (0, L) \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in [0, L] \\ \partial_x u(0, t) = h_0(t) & t \in [0, \infty) \\ \partial_x u(L, t) = h_L(t) & t \in [0, \infty) \end{cases}$$

where $h_0(0) = \Psi'(0)$ and $h_L(0) = \Psi'(L)$

Proposition 10.3.2 The solution to the heat equation in Neumann boundary condition is unique and is stable in the mean square sense.

10.4 Duhamel's Principle

Theorem 10.4.1 (Partial Solution to Heat Equation with Dirichlet Boundary Conditions) The heat equation with Dirichlet boundary conditions has solution of the form $u(x, t) = u_B(x, t) + u_I(x, t) + w(x, t)$ where

$$\begin{cases} h : [0, L] \rightarrow \mathbb{R} & \text{is smooth such that } h(0) = 0 \text{ and } h(L) = 1 \\ u_B(x, t) = g_0(t) + (g_L(t) - g_0(t))h(x) & (x, t) \in [0, L] \times [0, \infty) \\ u_I(x, t) = \Psi(x) - u_B(x, 0) & (x, t) \in [0, L] \times [0, \infty) \\ w(x, t) & (x, t) \in [0, L] \times [0, \infty) \end{cases}$$

where $w(x, t)$ satisfies the following homogenous boundary and initial values:

$$\begin{cases} \partial_t w(x, t) - k \partial_{xx} w(x, t) = f(x, t) & (x, t) \in [0, L] \times [0, \infty) \\ w(x, 0) = 0 & x \in [0, L] \\ w(0, t) = 0 & t \in [0, \infty) \\ w(L, t) = 0 & t \in [0, \infty) \end{cases}$$

where

$$f(x, t) = -(g'_0(t) + (g'_L(t) - g'_0(t))h(x)) + k(g_L(t) - g_0(t))h''(x) + k(\Psi''(x) - (g_L(0) - g_0(0)))h''(x)$$

Theorem 10.4.2 (Duhamel Principle) Let f be any function. Consider the following homogenous boundary and initial values differential equation:

$$\begin{cases} \partial_t w(x, t) - k \partial_{xx} w(x, t) = f(x, t) & (x, t) \in [0, L] \times [0, \infty) \\ w(x, 0) = 0 & x \in [0, L] \\ w(0, t) = 0 & t \in [0, \infty) \\ w(L, t) = 0 & t \in [0, \infty) \end{cases}$$

Let $v(x, t, \tau)$ denote a smooth solution to the following conditions with parameter τ :

$$\begin{cases} \partial_t v(x, t) = k \partial_{xx} v(x, t) & (x, t) \in (0, L) \times (\tau, \infty) \\ v(x, \tau) = f(x, \tau) & x \in [0, L] \\ v(0, t) = 0 & t \in [\tau, \infty) \\ v(L, t) = 0 & t \in [\tau, \infty) \end{cases}$$

Then

$$w(x, t) = \int_0^t v(x, t, \tau) d\tau$$

solves above homogenous boundary condition and initial values of the heat equation.

It now remains to solve the function v in Duhamel's principle, which is simply the heat equation and homogenous Dirichlet boundary conditions.

Proposition 10.4.3 Suppose that f has fourier series expansion as $f(x, 0) = \sum_{k=1}^n E_k \sin\left(\frac{k\pi}{L}x\right)$.

The solution to the following initial value problem

$$\begin{cases} \partial_t v(x, t) = k \partial_{xx} v(x, t) & (x, t) \in (0, L) \times (\tau, \infty) \\ v(x, \tau) = f(x, \tau) & x \in [0, L] \\ v(0, t) = 0 & t \in [\tau, \infty) \\ v(L, t) = 0 & t \in [\tau, \infty) \end{cases}$$

is given by

$$v(x, t) = \sum_{i=0}^{\infty} e^{-k\beta_i^2 t} D_i \sin(\beta_i x)$$

where $\beta_i = \frac{i\pi}{L}$ and $D_i =$

Proof WLOG take $\tau = 0$. We perform separation of variables again and assume that $v(x, t) = X(x)T(t)$. Substituting into the heat equation gives

$$\frac{T'(t)}{kT(t)} = \frac{X''(x)}{X(x)}$$

This ratio can only be constant since it cannot depend on x nor t . Similar to before we notice that $\lambda < 0$ thus let $\lambda = -\beta^2$. The solution to the ODE with X is given by

$$X(x) = A_k \cos(\beta x) + B_k \sin(\beta x)$$

Using the fact that $X(0) = 0$ and $X(\pi) = L$, we obtain

$$X_k(x) = D_k \sin(\beta_k x)$$

for $\beta_k = \frac{k\pi}{L}$ for $k \in \mathbb{N} \setminus \{0\}$. Now solving the ODE for T , we get

$$T_k(t) = C_k e^{-k\beta_k^2 t}$$

Now absorbing the coefficients C_k into D_k , we obtain the general solution

$$v(x, t) = \sum_{k=1}^{\infty} D_k e^{-k\beta_k^2 t} \sin\left(\frac{k\pi}{L} x\right)$$

Now when $t = 0$,

$$v(x, 0) = \sum_{k=1}^{\infty} D_k \sin\left(\frac{k\pi}{L} x\right) = f(x, 0)$$

Thus we can directly compare coefficients to get $D_k = E_k$. ■

Recalling all the steps we used, to solve an inhomogenous boundary conditions (the solution to Neumann is similar), we have the following procedure:

Step 1: Find u_B and u_I to get solutions with respect to the boundary conditions and initial value conditions.

Step 2: Calculate $f(x, t)$ from $w = u - u_B - u_I$ and the heat equation.

Step 3: Solve the homogenous boundary conditions problem to obtain v Step 4: Use Duhamel's principle to find $w(x, t) = \int_0^t v(x, t, \tau) d\tau$

10.5 Cauchy Problem and the Fundamental Solution

Definition 10.5.1 (Cauchy Problem of the Heat Equation) The Cauchy problem for the heat equation in one dimension consists of finding a function $u : \mathbb{R} \times (0, \infty) \rightarrow \mathbb{R}$ such that

$$\begin{cases} \partial_t u(x, t) = k \partial_{xx} u(x, t) & (x, t) \in \mathbb{R} \times (0, \infty) \\ u(x, 0) = \Psi(x) & x \in \mathbb{R} \end{cases}$$

Theorem 10.5.2 (Fundamental Solution of the Heat Equation) Let Ψ be continuous and bounded. Then

$$u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{4kt}} \Psi(y) dy$$

is smooth and solves the Cauchy problem of the heat equation.

11 Partial Differential Equation: Laplace's Equation

11.1 Laplace's Equation

Definition 11.1.1 (Laplacian Operator) Let $u : \mathbb{R}^2 \rightarrow \mathbb{R}$. Define the laplacian operator on u by

$$\Delta u(x, y) = \partial_{xx}u + \partial_{yy}u = \nabla \cdot \nabla u$$

Proposition 11.1.2 (Change of Variables) Using the substitution $x = r \cos(\theta)$ and $y = r \sin(\theta)$, we can change the laplacian operator into the form

$$\Delta u(x_1, x_2) = \Delta v(r, \theta) = \partial_{rr}u + \frac{1}{r^2} \partial_{\theta\theta}u + \frac{1}{r} \partial_r u$$

Theorem 11.1.3 The solution to the laplace's equation where u is a function of r and θ with the following conditions

$$\begin{cases} \partial_{rr}u + \frac{1}{r^2} \partial_{\theta\theta}u + \frac{1}{r} \partial_r u = 0 & u : (0, k) \times (0, \alpha) \\ u(k, \theta) = g(\theta) & \theta \in (0, \alpha) \\ u(r, 0) = u(r, \alpha) = 0 & r \in (0, k) \end{cases}$$

is given by

$$u(r, \theta) = \sum_{k=1}^{\infty} r^{\frac{k\pi}{\alpha}} A_k \sin\left(\frac{k\pi}{\alpha} \theta\right)$$

11.2 Fundamental Solution and Green's Function

Definition 11.2.1 (Poisson Problem) The poisson problem consists of finding a function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$-\Delta u = f$$

Theorem 11.2.2 (Fundamental Solution for the Laplacian) Let $f \in C^2(\mathbb{R}^n)$ be a function such that $\{x | f(x) \neq 0\}$ is bounded. Then the solution to the Poisson problem is given by

$$u(x_1, \dots, x_n) = \int_{\mathbb{R}^n} F(x - y) f(y) dy$$

where its second order partial derivatives are continuous and

$$F(x) = \begin{cases} -\frac{|x|}{2} & n = 1 \\ -\frac{1}{2\pi} \ln(|x|) & n = 2 \\ \frac{1}{n(n-2)V_n(B_1(0))} \cdot \frac{1}{|x|^{n-2}} & n \geq 3 \end{cases}$$

where $V_n(B_1(0))$ is the volume of the unit ball in n dimensions.

Definition 11.2.3 (Poisson Problem with Boundary Conditions) Let $\Omega \subseteq \mathbb{R}^n$ be an open bounded domain. Let $f : \Omega \rightarrow \mathbb{R}$ and $g : \partial\Omega \rightarrow \mathbb{R}$. The poisson problem with boundary conditions consists of finding a function $u : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\Delta u(\mathbf{x}) = f(\mathbf{x}) & \mathbf{x} \in \Omega \\ u(\mathbf{x}) = g(\mathbf{x}) & \mathbf{x} \in \partial\Omega \end{cases}$$

Theorem 11.2.4 Let $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ is harmonic. Then

$$\max_{x \in \Omega} u(x) = \max_{x \in \partial\Omega} \{u(x)\}$$

and

$$\min_{x \in \Omega} u(x) = \min_{x \in \partial\Omega} \{u(x)\}$$

The following principle gives the uniqueness and stability of our solution for the laplacian operator.

Theorem 11.2.5 (Dirichlet Principle) A function $u \in \mathcal{A} = \{v \in C^2(\partial\Omega) | v = g \text{ in } \partial\Omega\}$ solves $-\Delta u = f$ in Ω and $u = g$ in $\partial\Omega$ if and only if it minimizes I , meaning

$$I(u) = \min_{w \in \mathcal{A}} \{I(w)\}$$

where

$$I(w) = \int_{\Omega} \frac{1}{2} |\nabla w|^2 - fw \, dx$$